

6G Weekly Survey — Jul 13 - Jul 19, 2026

15 items across 5 topics, ranked by importance.

ISAC (sensing + communication)

Symbol-Level Precoding for Continuous-Aperture ISAC Systems

Hongli Liu, Qiang Li · arXiv eess.SP · 2026-07-10

★★★★■ Strong theoretical contribution addressing a fundamental challenge in continuous-aperture ISAC optimization with rigorous dimensional reduction result from a credible group, though CAPA technology maturity remains uncertain for 6G deployment.

This paper addresses symbol-level precoding for continuous-aperture arrays (CAPAs) in downlink ISAC systems, where optimizing continuous current distributions creates infinite-dimensional problems. The authors prove that optimal current lies in a finite-dimensional subspace spanned by communication and sensing electromagnetic responses, enabling exact reformulation and solution via penalty projected-gradient optimization. Simulations show improved sensing utility and communication reliability over Fourier-basis CAPA and discrete arrays.

Contribution: Establishing that optimal continuous current distributions for CAPA-ISAC lie in a finite-dimensional subspace defined by electromagnetic responses, transforming an infinite-dimensional optimization into a tractable finite-dimensional problem with structure preservation.

Relevance to your work: Directly relevant to your ISAC research interests, offering a mathematically principled framework for joint sensing-communication optimization in next-generation array architectures with symbol-level precoding that maximizes both functions simultaneously.

Fundamental Sensing Limits of 6G Cooperative MIMO-ISAC Networks: Joint Position-Velocity CRLB and Decoupling Analysis

Yanpeng Su, Norman Franchi, Maximilian Lübke · arXiv eess.SP · 2026-07-09

★★★★■ Provides fundamental performance bounds essential for ISAC system design with practical simplifications validated for optimization, addressing a gap in cooperative sensing theory relevant to 6G standardization.

This paper derives the exact Cramér-Rao lower bound (CRLB) for joint position-velocity estimation in cooperative MIMO-ISAC networks, accounting for coupling between position and velocity in the Fisher information matrix that prior work typically ignores. The authors derive full FIM expressions for multi-monostatic, multi-bistatic, and hybrid sensing configurations, then provide simplified decoupled approximations with validity criteria to enable tractable optimization. Simulations demonstrate the simplified CRLB introduces negligible error in general cases while offering substantially lower computational complexity.

Contribution: First rigorous treatment of position-velocity coupling in cooperative MIMO-ISAC CRLB analysis with practical decoupling criteria that enable tractable resource allocation and beamforming optimization.

Relevance to your work: Directly relevant to your ISAC research agenda, providing theoretical foundations for cooperative sensing performance limits and tractable bounds for AI-based resource allocation in multi-node ISAC architectures.

Low-Complexity Gridless Single-Snapshot DoA Estimation via Truncated Hankel Newton-MUSIC

Ruoxiao Cao, Wentao Yu, Yi Gong, Khaled B. Letaief · arXiv eess.SP · 2026-07-09

★★★★■ Addresses a critical practical bottleneck (single-snapshot sensing latency) for ISAC with large reconfigurable arrays, providing substantial computational savings with minimal accuracy loss—highly relevant for low-latency 6G sensing applications.

Proposes a low-complexity gridless DoA estimation method for single-snapshot scenarios in ISAC systems with reconfigurable antenna arrays. Uses truncated (non-square) Hankel matrix construction to reduce computational cost from quadratic to linear in array size, combined with Newton refinement to eliminate grid quantization errors. Achieves near-optimal accuracy with over 100× runtime reduction for large arrays compared to conventional Hankel-MUSIC.

Contribution: Truncated Hankel matrix construction with fixed row dimension achieves linear complexity scaling while preserving DoA estimation accuracy, enabling real-time sensing for large reconfigurable arrays in dynamic ISAC scenarios.

Relevance to your work: Directly applicable to ISAC beam management and dynamic channel sensing in reconfigurable intelligent surface (RIS) or large-scale MIMO ISAC systems where low-latency single-snapshot DoA estimation is essential.

Beyond Single-Band: Analysis and Resource Allocation for Multi-band ISAC Systems

Haotian Liu, Zhiqing Wei, Xingwang Li, Yunxin Geng, et al. · arXiv eess.SP · 2026-07-09

★★★★■ Addresses a practical 6G deployment constraint (fragmented spectrum) with rigorous analytical tools and substantial performance gains, directly relevant to standards discussions on ISAC waveform design.

This paper addresses multi-band ISAC for 6G, where spectrum fragmentation motivates using multiple separated frequency bands to improve sensing robustness against frequency-selective target scattering. The authors derive closed-form detection/false-alarm probabilities for multi-band OFDM radar signals and propose an ADMM-based joint power and time-frequency resource allocation scheme. Simulations show 18 dB detection gain over single-band baselines at 90% detection probability.

Contribution: First analytical characterization of detection statistics for multi-band OFDM ISAC signals with joint power and time-frequency resource optimization across non-contiguous bands.

Relevance to your work: Directly relevant: multi-band operation is critical for realistic ISAC deployments in fragmented 6G spectrum, and the analytical framework for joint sensing-communication resource allocation complements your work on AI-native air interfaces for ISAC.

Fused Constrained Policy Reuse Optimization for Wireless Resource Allocation

An Liu, Zheyuan Zhou, Kexuan Wang · arXiv eess.SP · 2026-07-10

★★★★■ Solid work addressing the practical challenge of safe, sample-efficient DRL in wireless systems with theoretical grounding, though limited to simulation validation on known problem settings without experimental data or commercial deployment insights.

This paper proposes Fused-CPRO, a constrained deep reinforcement learning method for wireless resource allocation that fuses a learnable target policy with source policies from related scenarios and domain-knowledge expert rules. The method uses constrained stochastic successive convex approximation to optimize policy mixing weights and target parameters under long-term constraints, with convergence guarantees to KKT points. Simulations on MU-MIMO power control and MIMO-ISAC beamforming show faster convergence and better constraint satisfaction than baseline methods.

Contribution: The novel fusion framework combines heterogeneous policy priors (learned, transferred, and expert) with formal constrained optimization and provable convergence, enabling safe online learning with minimal environment interactions for resource allocation under stringent constraints.

Relevance to your work: Directly relevant to your ISAC work: the paper demonstrates Fused-CPRO on MIMO-ISAC beamforming with Cramer-Rao bound constraints for sensing performance, showing how constrained policy learning can balance communication and sensing requirements in integrated systems.

Design and Deployment Guidelines for UAV-Mounted RIS Under Position Uncertainty

Kevin Weinberger, David Müller, Martin Mönnigmann, Aydin Sezgin · arXiv eess.SP · 2026-07-08

★★★★ Solid contribution addressing a practical UAV-RIS deployment challenge with validated analytical tools, though impact depends on whether industry pursues aerial RIS versus terrestrial alternatives.

This paper analyzes how UAV position uncertainty degrades phase coherence in UAV-mounted reconfigurable intelligent surfaces (RIS) for 6G networks. Using a GUM-based uncertainty propagation framework, the authors derive closed-form models showing phase uncertainty causes exponential coherence loss and introduce a performance-driven coherence threshold (PCT) to define deployment boundaries. Results demonstrate that positioning uncertainty forces optimal UAV-RIS placement away from conventional near-Tx or near-Rx locations, with Monte Carlo validation confirming the analytical model.

Contribution: First rigorous stochastic channel model quantifying how UAV position uncertainty propagates to phase misalignment in cascaded RIS channels, yielding practical deployment guidelines that contradict static RIS placement intuition.

Relevance to your work: Directly relevant to ISAC and 6G deployment scenarios where RIS mobility could support dynamic beam management and localization, though your work focuses more on learned PHY and channel modeling than hardware deployment constraints.

AI-native air interface

Generative Communications: Overview, Technologies, and Trends

Wenjun Zhang, Zhiyong Chen, Tong Wu, Guo Lu, et al. · arXiv cs.IT · 2026-07-10

★★★★ Strong conceptual framework from credible research group that aligns with emerging 6G vision of task-oriented and semantic communication, though practical viability and standardization path remain to be demonstrated.

This paper introduces generative communications (GenCom) as a 6G paradigm where large AI models enable transmitters to send minimal information while receivers use shared generative priors to synthesize intended content, fundamentally shifting from data reproduction to controlled generation. The authors propose a two-layer architecture with enabling technologies and analyze four application scenarios demonstrating ultra-efficient transmission and semantic-level robustness. The work formalizes GenCom's AI-native and generation-driven properties as foundational concepts for next-generation networks.

Contribution: First systematic formalization of generative communications as a core 6G paradigm, establishing theoretical foundations for AI-driven content synthesis at the receiver rather than traditional bit-accurate transmission.

Relevance to your work: Directly relevant to your AI-native air interface research as it provides architectural foundations for integrating generative models into the communication stack, complementing your work on semantic communication and learned physical layer design.

Channel modeling / GAN-based

Rydberg Atomic Quantum Radio: A Comprehensive Survey From Wireless Communication Perspective

Yiyue Xiang, Neng Ye, Qihao Peng, Junrui Zhao, et al. · arXiv eess.SP · 2026-07-07

★★★★ Well-motivated survey of an emerging receiver technology with potential long-term impact, but RAQ radios remain early-stage laboratory demonstrations far from commercial deployment timelines relevant to 6G standardization. This survey reviews Rydberg atomic quantum (RAQ) radios—quantum-enabled RF receivers that map electromagnetic fields to atomic quantum states—from a wireless communications perspective, covering underlying quantum mechanisms, architectures, atomic response models, and performance trade-offs among sensitivity, bandwidth, and operating frequency. The authors systematically analyze equivalent channel modeling frameworks for RAQ systems and examine applications including cognitive radio, MIMO, satellite networks, and integrated sensing and communication. The work aims to bridge atomic physics and wireless communications to address full-spectrum agility, ultra-high sensitivity, and anti-jamming requirements of next-generation space-air-ground-sea integrated networks.

Contribution: First comprehensive survey connecting Rydberg atomic quantum receiver physics with wireless communication theory through equivalent channel models and application scenarios relevant to 6G systems including ISAC and satellite networks.

Relevance to your work: Directly relevant to ISAC research and NTN/satellite scenarios; the equivalent channel modeling framework could inform future work on non-conventional receiver architectures, though practical deployment remains distant.

Modeling and Analysis for Multiple-Layer LEO Satellite Internet of Things Constellations

Ming Ying, Xiaoming Chen, Qiao Qi, Yichao Xu · arXiv cs.IT · 2026-07-10

★★★★■ Solid theoretical contribution for NTN system-level analysis with practical channel modeling, though focused on IoT narrowband scenarios rather than broadband 6G use cases.

This paper develops a stochastic geometry framework for analyzing multiple-layer LEO satellite IoT constellations under practical Rician fading channels using Cox point processes. A novel channel approximation method enables closed-form expressions for connectivity, coverage probability, and transmission rate in short-packet scenarios. Simulations validate the model and provide deployment insights for multi-layer satellite architectures.

Contribution: First tractable analytical framework combining stochastic geometry, Cox processes, and Rician fading for multi-altitude LEO constellations with closed-form performance metrics for IoT short-packet transmission.

Relevance to your work: Directly relevant to your NTN research interests by providing rigorous channel modeling and analytical tools for multi-layer LEO architectures that could inform direct-to-device 6G designs beyond IoT.

6G Demos & Trials

Small Language Model-based Control for BBR over Low Earth Orbit Satellite Internet

Rakshitha De Silva, Shiva Raj Pokhrel, Jonathan Kua · arXiv cs.NI · 2026-07-08

★★★★■ Solid empirical work addressing a real LEO transport-layer problem with global Starlink measurements, though the SLM control approach is more engineering optimization than fundamental wireless innovation.

This paper evaluates BBR congestion control over a global SpaceX Starlink testbed, finding that while BBR achieves superior throughput compared to Cubic, Vegas, and Hybla, it incurs excessive retransmissions due to aggressive bandwidth probing over LEO links. To mitigate this throughput-stability trade-off, the authors propose an SLM-guided BBR adaptation framework using lightweight language models (GPT-2, T5, GPT-Neo, SmoLM2) with LoRA fine-tuning to learn phase-safe pacing-gain decisions from real Starlink traces. The SLM approach retains BBR's throughput advantage while substantially reducing retransmissions at low computational cost.

Contribution: First use of small language models for real-time congestion control adaptation in LEO satellite networks, with experimental validation on operational Starlink infrastructure across six AWS endpoints.

Relevance to your work: Directly relevant to NTN research priorities; demonstrates practical challenges in LEO satellite transport that may inform cross-layer design for non-terrestrial air interfaces and latency-sensitive 6G services.

Unveiling TCP BBR Dominance in Starlink Internet: Experimental Insights and Analysis

Rakshitha De Silva, Shiva Raj Pokhrel, Jonathan Kua · arXiv cs.NI · 2026-07-08

★★★★■ Solid empirical contribution for LEO satellite transport optimization, relevant to NTN standardization, but focuses on transport layer rather than physical/MAC layer design for 6G.

This paper experimentally evaluates Google's BBR-v3 congestion control algorithm across SpaceX's Starlink LEO satellite network using a six-city testbed spanning five continents, benchmarking it against eight competing algorithms (Cubic, Hybla, Vegas, LeoCC, Copa, PCC, BBR-v1, BBR-v2). BBR-v3 demonstrates superior balanced performance in fairness-loss-delay tradeoffs rather than raw throughput maximization, particularly under high-latency variable satellite conditions. The authors develop mathematical models characterizing BBR-v3 behavior and Starlink network dynamics to explain the empirical observations.

Contribution: First large-scale global experimental validation of BBR-v3 over operational Starlink infrastructure with pragmatic models explaining why BBR-v3 achieves balanced fairness and throughput in LEO satellite environments.

Relevance to your work: tangential - addresses LEO/NTN transport protocols rather than air interface design, channel modeling, or physical layer AI/ML that constitutes your core research focus

Robustifying Vision-Language Models via Test-Time Prompt Adaptation

Xingyu Zhu, Huanshen Wu, Shuo Wang, Beier Zhu, et al. · arXiv cs.LG · 2026-07-10

★☆☆☆☆ This is computer vision and adversarial ML work with no connection to wireless communications, physical layer design, or 6G research despite the topic-matching error.

This paper addresses adversarial robustness in vision-language models (VLMs) like CLIP by proposing RITA, a test-time prompt adaptation method that uses optimal transport to align distributions of augmented visual features with text prototypes rather than relying on sample-level confidence. The approach maintains a dynamic cache of reliable test samples for online refinement, mitigating adversarial perturbations while preserving clean accuracy. Experiments show improved robustness against adversarial attacks compared to existing test-time adaptation methods.

Contribution: Novel use of optimal transport for distribution-level cross-modal alignment during test-time adaptation, shifting away from brittle sample-level confidence heuristics to improve adversarial robustness in VLMs.

Relevance to your work: tangential

Dissipativity-Based Multiport Stability Root-Cause Identification and Mitigation for Solid-State Transformers

Xiangyu Meng, Dong Xie, Hongjian Lin, Chunxu Lin, et al. · arXiv eess.SY · 2026-07-10

★☆☆☆☆ This work is entirely focused on power electronics and grid stability for solid-state transformers, with no connection to wireless communication, physical layer processing, or 6G air interface technologies.

This paper addresses low-frequency oscillation instability in solid-state transformers (SSTs) under weak grid conditions by establishing a multiport admittance matrix model covering AC dq-axes and DC ports. Through dissipativity analysis, the authors identify that coupling-dissipativity failures from synchronization loop dynamics—not localized self-dissipativity issues—are the dominant instability mechanism. They propose a dynamics-free orthogonal signal reconstruction controller that reshapes admittance characteristics to mitigate the identified coupling deficiencies, validated on a down-scaled SST prototype.

Contribution: The novel decomposition of passivity conditions into self- and coupling-dissipativity indices enables precise root-cause diagnosis of multiport instability in SSTs, revealing that synchronization loop coupling dynamics dominate over localized port issues.

Relevance to your work: tangential

Structural Decoupling and Current-Angle Steering for Post-Fault Recovery of Current-Limited Grid-Forming Inverters

Neethu Sajeev, Stephen Arinze Obi, Jae-Jung Jung · arXiv eess.SY · 2026-07-10

★☆☆☆☆ This is power electronics grid integration work focused on fault recovery in utility-scale inverters, with no connection to wireless communication systems or 6G physical layer research.

This paper addresses post-fault recovery failures in grid-forming inverters under current limiting, where conventional PI voltage control creates unstable recovery boundaries leading to trapping or oscillations. The authors propose a framework combining structurally decoupled virtual admittance voltage control with current-angle steering to stabilize the recovery boundary. Experimental validation on a 3-kVA prototype demonstrates reliable fault recovery under symmetric and asymmetric voltage sags.

Contribution: Identifies the root cause of post-fault recovery failures as interaction between voltage controller and current limiter creating moving recovery boundaries, and proposes a decoupled control architecture that stabilizes this boundary.

Relevance to your work: tangential

6G Physical Layer

Multiuser Zak-OTFS on the Uplink with Superimposed Spread-Pilots

Sai Pradeep Muppaneni, Ananthanarayanan Chockalingam · arXiv cs.IT · 2026-07-09

★★★★ Solid theoretical contribution to Zak-OTFS multiuser access with practical pilot design, though limited to specific pulse shapes and TF-shift orthogonality assumptions that may constrain deployment scenarios.

This paper addresses uplink multiuser Zak-OTFS with heterogeneous delay-Doppler frames by assigning users non-overlapping time-frequency shifts, deriving closed-form effective DD channel expressions for sinc and Gaussian pulses. A superimposed spread-pilot scheme based on FFT-transformed Zadoff-Chu sequences enables channel estimation through iterative IOR estimation and data detection, achieving negligible inter-user interference. Simulations show performance matching single-user systems, with sinc pulses achieving better spectral efficiency than embedded pilots across varying power ratios, while Gaussian pulses favor embedded pilots due to residual interference.

Contribution: Closed-form DD channel characterization for multiuser Zak-OTFS with TF-shift-based multiple access and a superimposed spread-pilot framework that decouples multiuser estimation into single-user problems with negligible IUI.

Relevance to your work: Directly relevant to your 6G waveform research (OTFS variants) and pilot design for AI-native air interfaces, offering analytical foundations for multiuser DD systems that could inform neural receiver architectures.